

Forecasting the U.S. Monthly Unemployment Rate

ECO 5740 - Forecasting and Time Series Models

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1 Abstract

This report presents a time series analysis and forecast of the U.S. Monthly Unemployment Rate (Not Seasonally Adjusted), series UNRATENSA, using data sourced from the Federal Reserve Economic Data (FRED) database. The dataset spans January 2015 through September 2025, providing 129 monthly observations. The unemployment rate is a critical indicator of labor market health, and accurate short-term forecasts are valuable to a wide range of stakeholders including policymakers, businesses, and job seekers. Following exploratory analysis and classical decomposition, a Box-Cox transformation was applied to stabilize variance across the series. Twelve individual forecasting models and four ensemble forecasting models were evaluated on a 26-month test set. The ensemble combining Holt-Winters' Additive Damped with Additive Seasonality (Holt_Winters_Ad_A) and the Theta method (Theta_A), labeled comb2, produced one of the lowest RMSEs among the ensemble models and passed residual diagnostics with a Ljung-Box p-value of 1.000, indicating the residuals are consistent with white noise. The final comb2 forecast projects the U.S. unemployment rate to exhibit a modest upward drift from approximately 4.3% toward 5.0% over the next 26 months, driven by the seasonal and trend components learned by the Holt-Winters' component. These findings carry important implications for Federal Reserve rate-setting, business hiring strategy, and students entering the labor market in the near future.

2 Introduction

The unemployment rate is one of the most closely watched macroeconomic indicators in the United States. It measures the percentage of the civilian labor force that is actively seeking but unable to find employment. This provides a strong indicator of economic health. When unemployment rises, consumer spending typically contracts, tax revenues fall, and pressure on fiscal and monetary policy increases. Conversely, declining unemployment is associated with stronger household balance sheets, rising wages, and broader economic expansion.

Forecasting the unemployment rate accurately is therefore has direct consequences for a wide range of stakeholders. The Federal Reserve monitors unemployment alongside price stability, using it to make interest rate decisions. State and local governments rely on unemployment trends to plan workforce development programs and budget unemployment insurance expenditures. Businesses use labor market projections to guide hiring plans and compensation decisions. Students and recent graduates find labor market conditions relevant when evaluating job prospects and timing career transitions.

This report addresses the following forecasting question. Given the trajectory of the U.S. unemployment rate from January 2015 through September 2025, which forecasting model best predicts its behavior over the next 26 months, and what does that forecast imply for future labor market conditions?

The dataset used is the U.S. Monthly Unemployment Rate, series UNRATENSA, from the Bureau of Labor Statistics via FRED. The series spans January 2015 to September 2025 and contains 129 monthly observations. Monthly frequency is appropriate for capturing short-run labor market dynamics that not be seen as well at a quarterly level. The not-seasonally-adjusted series is used deliberately: preserving the seasonal variation allows it to be explicitly modeled, which is consistent with the decomposition and modeling techniques applied in this course.

The unemployment rate has been extensively studied in the forecasting literature. Classical ARIMA models have long been applied to labor market time series, while more recent work has explored exponential smoothing ensembles and machine learning approaches. A recurring finding is that simple benchmark models are difficult to outperform at short horizons, particularly during periods of labor market stability. The COVID-19 pandemic represents the most dramatic structural break in recent history. In April 2020, the U.S. unemployment rate surged to 14.4% before recovering rapidly over the following two years. Any model trained on data spanning this episode may assign undue weight to this outlier, complicating out-of-sample forecast accuracy.

The comb2 ensemble forecast projects the U.S. unemployment rate to exhibit a modest upward drift from approximately 4.3% toward 5.0% over the next 26 months, with the characteristic seasonal pattern of winter peaks and summer troughs, and widening prediction intervals reflecting growing uncertainty at longer horizons.

For policymakers at the Federal Reserve, a gradually rising unemployment outlook suggests monitoring for signs of labor market softening, while still supporting a data-dependent approach to rate decisions. For businesses, the forecast implies a labor market that may become incrementally less competitive over the horizon, though conditions remain broadly stable in the near term. For students and recent graduates, market conditions appear accessible over the forecast horizon, with no dramatic deterioration anticipated in the near term.

3 Data Description and Sources

This project uses U.S. Monthly Unemployment Rate data from the U.S. Bureau of Labor Statistics, accessed via the Federal Reserve Bank of St. Louis (FRED) database. The unemployment rate is calculated as the percentage of the civilian labor force that is unemployed. It is defined as individuals who are jobless, available for work, and actively seeking employment. The version of the data used is not seasonally adjusted, which preserves the seasonal variation in monthly unemployment for modeling.

The series spans January 2015 to September 2025, providing 129 monthly observations. The series begins at 6.1% in January 2015 and ends at 4.3% in September 2025, with a full-sample mean of approximately 4.66%. A pronounced outlier occurs in April 2020, when the rate surged to 14.4% due to widespread business closures and layoffs associated with the COVID-19 pandemic. Outside of this episode, the series trended steadily downward from 2015 through early 2020, reaching a pre-pandemic low of 3.3%, before recovering and stabilizing in the 3.5%–4.6% range through the end of the sample. The dataset contains no missing values and required no cleaning prior to analysis.

4 Time Series Graphics

4.1 Set up data as a time series object

Below, I have brought the data into R and set it up as a time series object. I have called `read_csv()` to import the data, `mutate()` to create a Month index using `yearmonth()`, and `as_tsibble()` to convert the data frame into a tsibble time series object indexed by month.

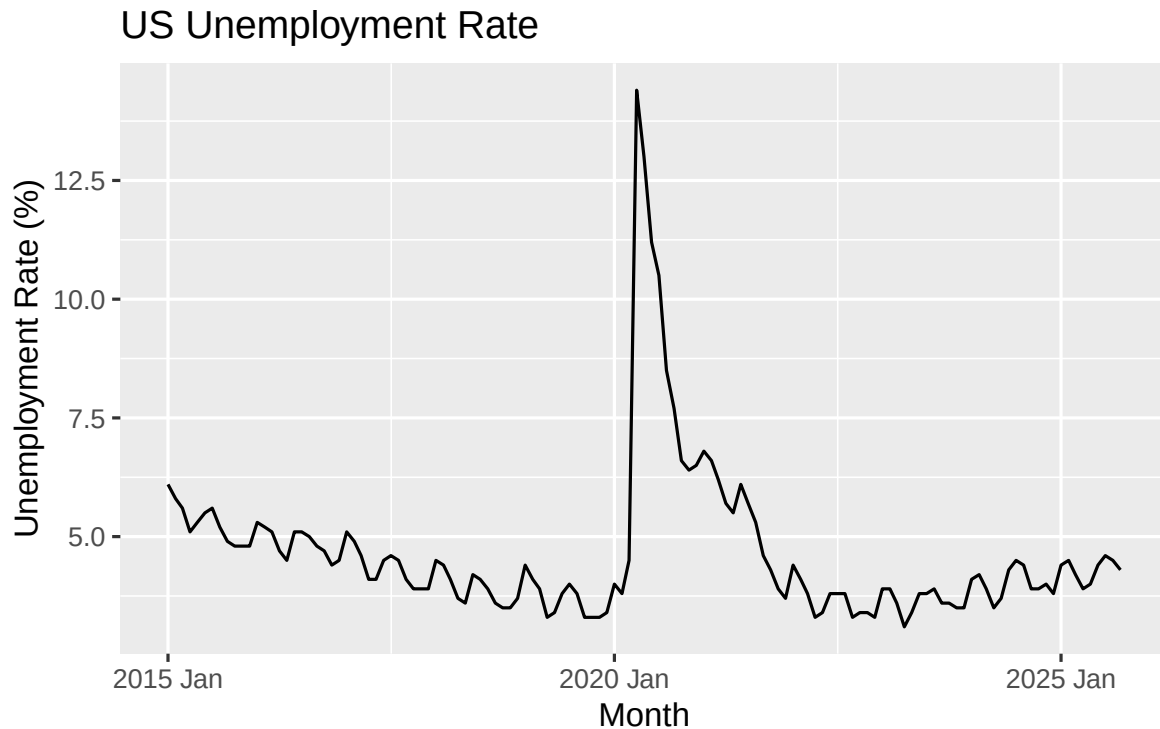
```
mydata <- read_csv("C:/Users/waklu/OneDrive/Desktop/ECON5740/UNRATENSA.csv")

mydata <- mydata |>
  mutate(Month = yearmonth(observation_date)) |>
  as_tsibble(index = Month)
```

4.2 Plot the data

Below, I have plotted the data using `autoplot()`. The series displays a clear downward trend from January 2015 through early 2020, reflecting a sustained period of economic expansion. In March and April 2020, the rate spikes dramatically to 14.4%, driven by the sudden economic disruption caused by the COVID-19 pandemic. From May 2020 onward, the rate recovers rapidly, returning to pre-pandemic levels by late 2022. From late 2022 through September 2025, the series exhibits a modest upward trend, stabilizing in the 3.5%–4.6% range.

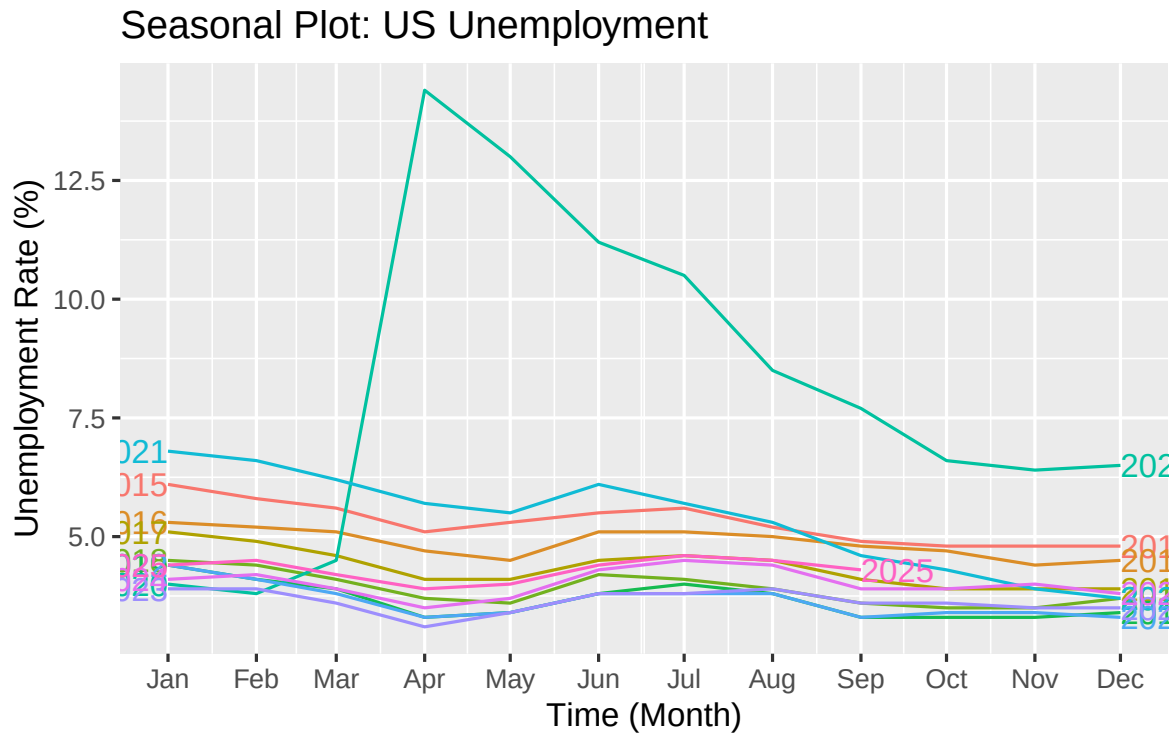
```
mydata |>
  autoplot(UNRATENSA) + ggtitle("US Unemployment Rate") + xlab("Month") + ylab("Unemployment Rate")
```



4.3 Time series patterns

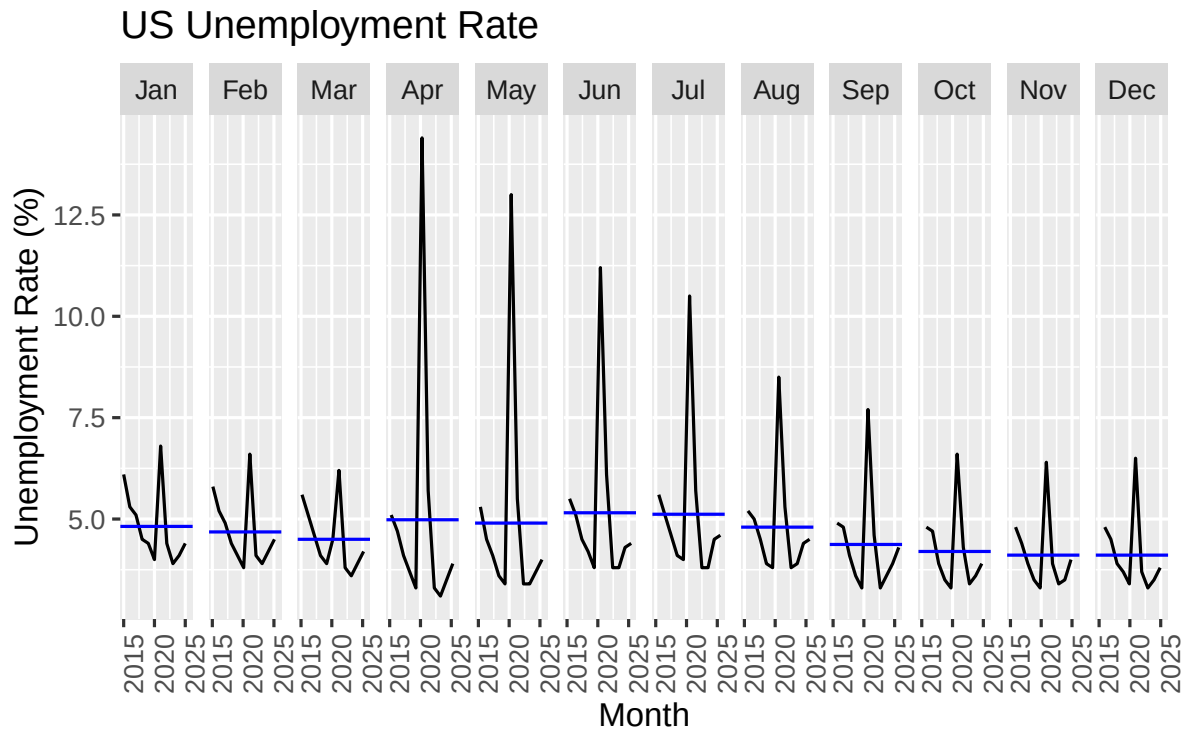
The seasonal plot below shows each calendar year as a separate line, with the month of the year on the x-axis. Each year follows a broadly similar pattern. Unemployment tends to be higher in January and February, dips through the spring and early summer months, and rises again slightly in the late summer and fall. This is consistent with seasonal labor supply dynamics such as school graduations and industry seasonality. The teal line for 2020 is clearly visible as an outlier, surging through April before declining through the rest of the year.

```
gg_season(mydata, UNRATENSA, labels = "both") + labs(title = "Seasonal Plot: US Unemployment",  
  y = "Unemployment Rate (%)", x = "Time (Month)")
```



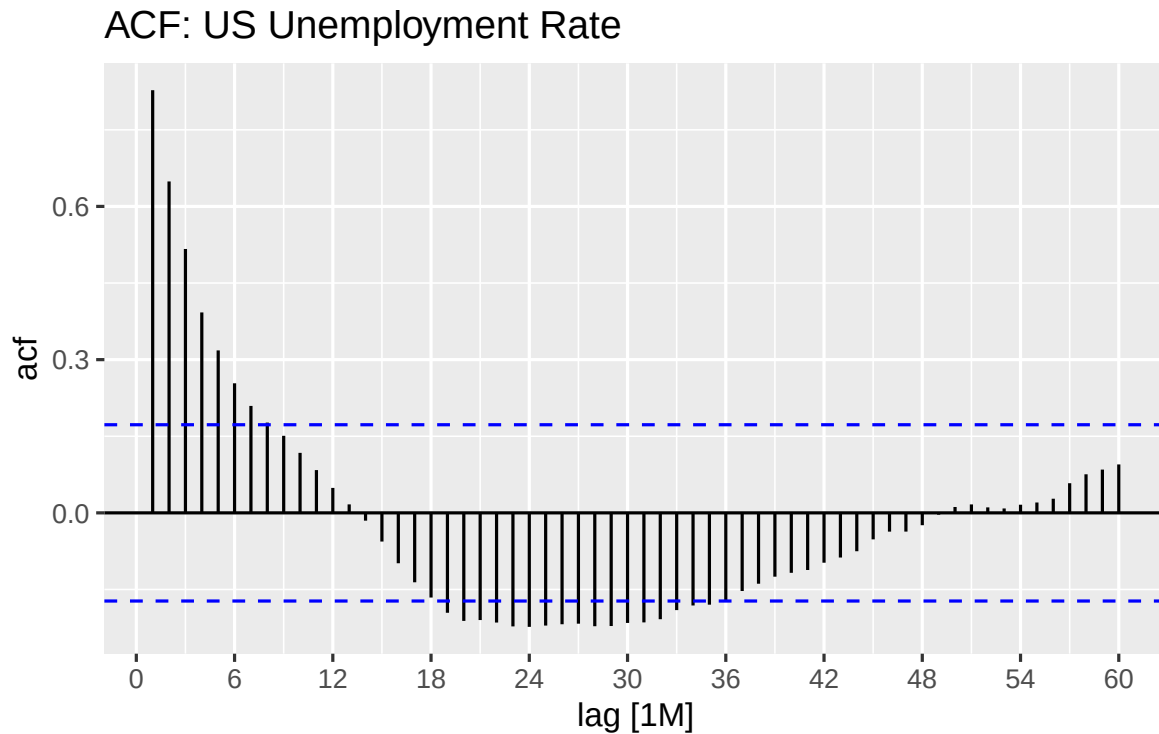
The subseries plot below reinforces this finding by showing each month's time series in a separate panel, with the blue horizontal line indicating the long-run average for that month. The averages in the subseries are skewed upward for some months due to the outlier in 2020. The averages are higher for summer months. However, the winter months have higher values for the most part outside of 2020.

```
gg_subseries(mydata, UNRATENSA) + labs(title = "US Unemployment Rate", y = "Unemployment Rate")
```



The ACF plot below examines autocorrelation at lags up to 60 months. The ACF declines slowly from lag 1, with significant positive autocorrelation persisting through approximately lag 8 before crossing the significance line. Negative autocorrelations then appear at lags 19–36, reflecting a reversal pattern consistent with the post-COVID recovery. After lag 36, autocorrelations are small and mostly within the bounds.

```
ACF(mydata, UNRATENSA, lag_max = 60) |>
  autoplot() + labs(title = "ACF: US Unemployment Rate")
```



A distinct downward trend is evident in the data from 2015 to 2020, indicating a sustained strengthening of the labor market. The ACF's slow decay confirms this trend component. The seasonal plot and subseries plot together confirm a regular seasonal pattern, with January and February as the peak unemployment months. There is also a cyclical component. The rate rises and falls at irregular intervals driven by economic and policy developments, most dramatically illustrated by the COVID-19 shock. The April 2020 value of 14.4% is a clear unusual observation.

5 Transformations and Adjustments

5.1 Calendar Adjustment

A calendar adjustment was not applied to this series. The unemployment rate is a percentage of the civilian labor force rather than a count or dollar-denominated flow, so it is not directly affected by the varying number of working days across months.

5.2 Population Adjustment

A population adjustment was not applied. The unemployment rate is already expressed as a share of the civilian labor force, which controls for changes in population size over time. A further adjustment would be unnecessary.

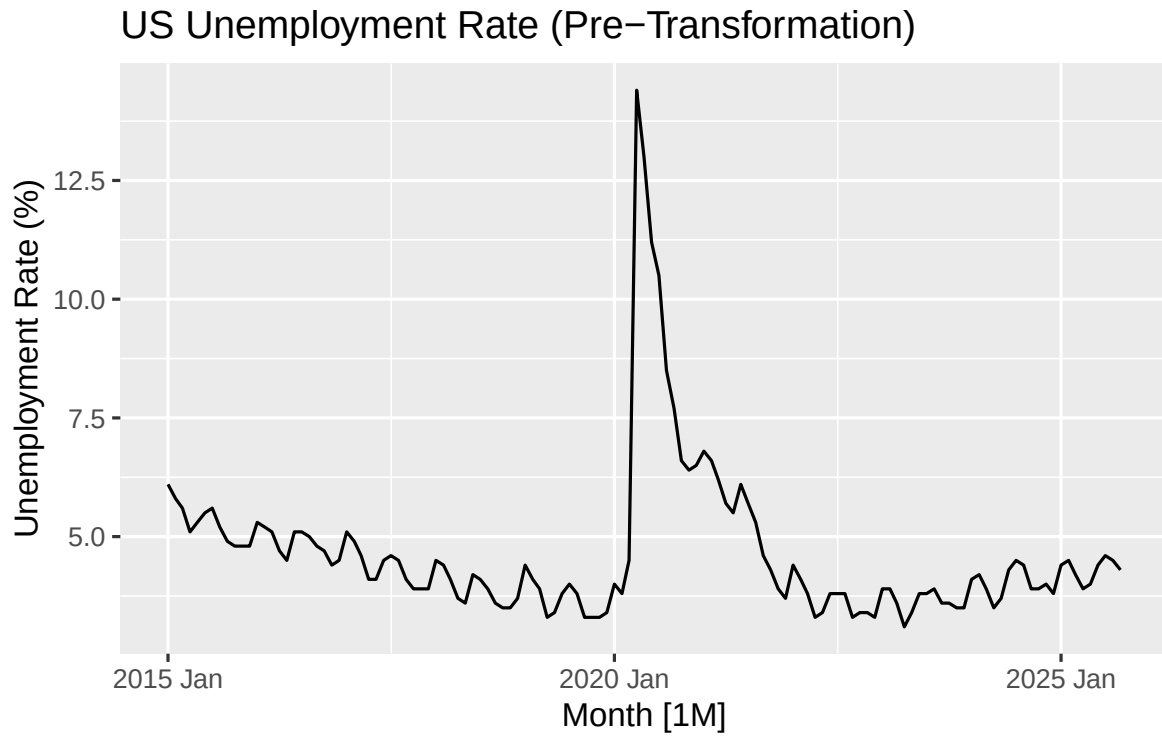
5.3 Inflation Adjustment

An inflation adjustment was not applied. The unemployment rate is a unitless ratio and carries no dollar denomination, so inflation does not affect its measurement.

5.4 Mathematical Transformation

A Box-Cox transformation was applied to stabilize the variance of the series. The optimal lambda value was determined using the Guerrero method via the `features()` function. As shown in the plot below, the transformation $\lambda = -0.9$ compresses the large COVID spike and produces a series with more homogeneous variance, which is beneficial for model fitting.

```
mydata |>
  autoplot(UNRATENSA) + labs(title = "US Unemployment Rate (Pre-Transformation)",
    y = "Unemployment Rate (%)", x = "Month [1M]")
```



```
lambda <- mydata |>
  features(UNRATENSA, features = guerrero) |>
  pull(lambda_guerrero)

mydata |>
  autoplot(box_cox(UNRATENSA, lambda)) + labs(y = "", title = latex2exp::TeX(paste0("Trans",
  round(lambda, 2))))
```



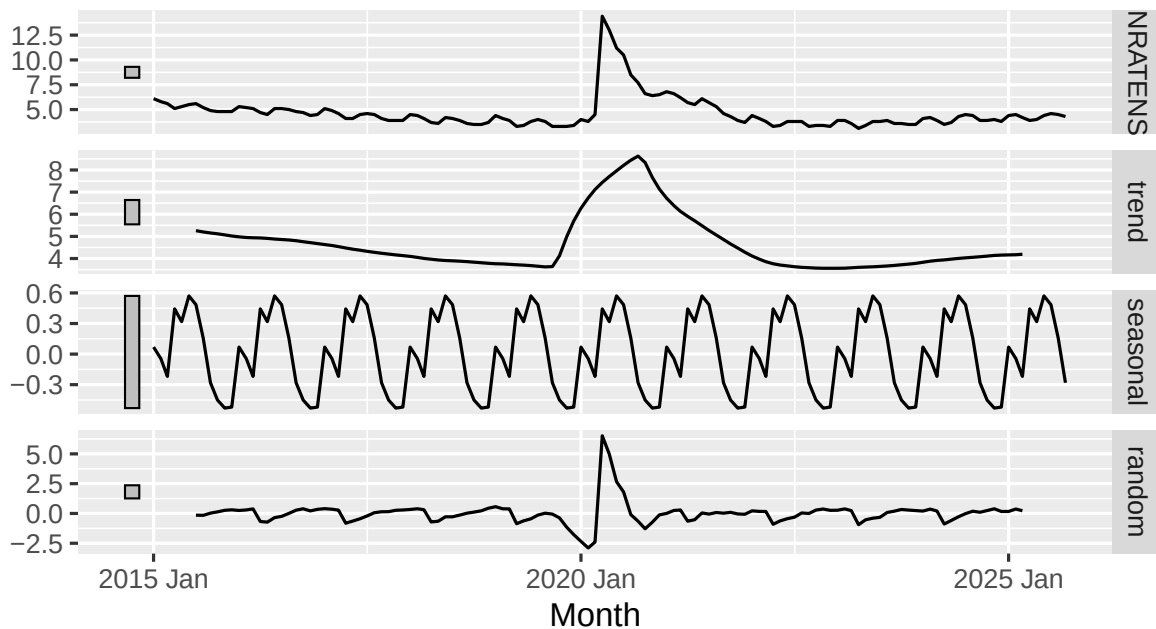
6 Time Series Decomposition

In the figure below, the time series has been broken down into three components using classical additive decomposition. The top panel shows the complete raw data. The second panel, trend, shows the long-run movement in the series. There is a steady decline from 2015 through early 2020, the sharp COVID spike, the subsequent recovery, and a stabilization after. The third panel, seasonal, reveals the regular within-year pattern. Unemployment peaks in January and troughs in summer. This confirms the seasonality identified in the seasonal and subseries plots above. The fourth panel, random, captures the unexplained variation. It is mostly small and centered near zero, with the exception of the large positive residual in April–May 2020, which no trend or seasonal model can anticipate from historical data alone. When the bottom three panels are combined, they reproduce the original series shown in the top panel.

```
mydata |>
  model(classical_decomposition(UNRATENSA, type = "additive")) |>
  components() |>
  autoplot() + labs(title = "Classical Additive Decomposition of US Unemployment Rate")
```

Classical Additive Decomposition of US Unemployment Rate

$$\text{UNRATENSA} = \text{trend} + \text{seasonal} + \text{random}$$



7 Forecasting Methods

7.1 Selecting a Reliable Forecasting Model

7.1.1 Training set

There are 129 total observations in this dataset, spanning January 2015 through September 2025. Following the standard 80/20 split, the first 103 observations (January 2015 through July 2023) are assigned to the training set, and the remaining 26 observations (August 2023 through September 2025) form the test set. The forecast horizon h is set equal to the length of the test set.

```
training_set <- mydata |>
  filter_index("2015 Jan" ~ "2023 Jul")

test_set <- mydata |>
  filter_index("2023 Aug" ~ "2025 Sep")
```

```
h <- nrow(test_set)
```

7.1.2 Fit the model

Twelve individual forecasting models are fit to the Box-Cox-transformed unemployment rate using the training set. The model set spans a wide range of complexity: four benchmark methods (Mean, Naive, Drift, Seasonal Naive) included for comparison purposes, two Holt-Winters' exponential smoothing specifications with damped trend (additive and multiplicative seasonality), an auto-selected ETS, an auto-selected SARIMA, a neural network autoregression, a Theta model, and two Prophet models (additive and multiplicative seasonality). Four ensemble combinations of the top three advanced models are also constructed to evaluate whether combining forecasts improves accuracy.

Additive seasonality is included as the primary choice for both Holt-Winters' and Theta, since the seasonal swings in UNRATENSA appear roughly constant in size across different rate levels. The damped trend is used in both Holt-Winters' specifications to avoid unrealistic long-run projections, pulling the forecast toward a flat trajectory appropriate for a series stabilizing near equilibrium. SARIMA is included as a statistically principled model that selects its own differencing and lag structure directly from the data — the auto-selected specification returned ARIMA(1,0,1)(1,0,0),12, confirming a seasonal component at lag 12. N_Net and Prophet round out the set as flexible, data-driven methods.

```
set.seed(123)
my_fit <- training_set |>
  model(Mean = MEAN(box_cox(UNRATENSA, lambda)), Naive = NAIVE(box_cox(UNRATENSA,
    lambda)), Drift = NAIVE(box_cox(UNRATENSA, lambda) ~ drift()), Seasonal_naive = SNAI
    lambda)), Holt_Winters_Ad_A = ETS(box_cox(UNRATENSA, lambda) ~ error("A") +
    trend("Ad") + season("A")), Holt_Winters_Ad_M = ETS(box_cox(UNRATENSA, lambda) ~
    error("A") + trend("Ad") + season("M")), ETS_auto = ETS(box_cox(UNRATENSA,
    lambda)), SARIMA = ARIMA(box_cox(UNRATENSA, lambda), stepwise = FALSE, approx = FALS
    N_Net = NNETAR(box_cox(UNRATENSA, lambda)), Theta_A = THETA(box_cox(UNRATENSA,
    lambda) ~ season(period = 12)), Prophet_A = prophet(box_cox(UNRATENSA,
    lambda) ~ season(period = 12, order = 2, type = "additive")), Prophet_M = prophet
    lambda) ~ season(period = 12, order = 2, type = "multiplicative")) |>
  mutate(comb1 = (Holt_Winters_Ad_A + Holt_Winters_Ad_M)/2) |>
  mutate(comb2 = (Holt_Winters_Ad_A + Theta_A)/2) |>
  mutate(comb3 = (Holt_Winters_Ad_M + Theta_A)/2) |>
  mutate(comb4 = (Holt_Winters_Ad_A + Holt_Winters_Ad_M + Theta_A)/3)
```

7.1.3 Compare the forecast accuracy across models

Each model is used to produce a 26-period forecast over the test set. Accuracy is assessed using RMSE, MASE, and RMSSE, arranged in ascending order of RMSE so that the best-performing models appear at the top.

```
my_fc <- my_fit |>
  forecast(h = h)

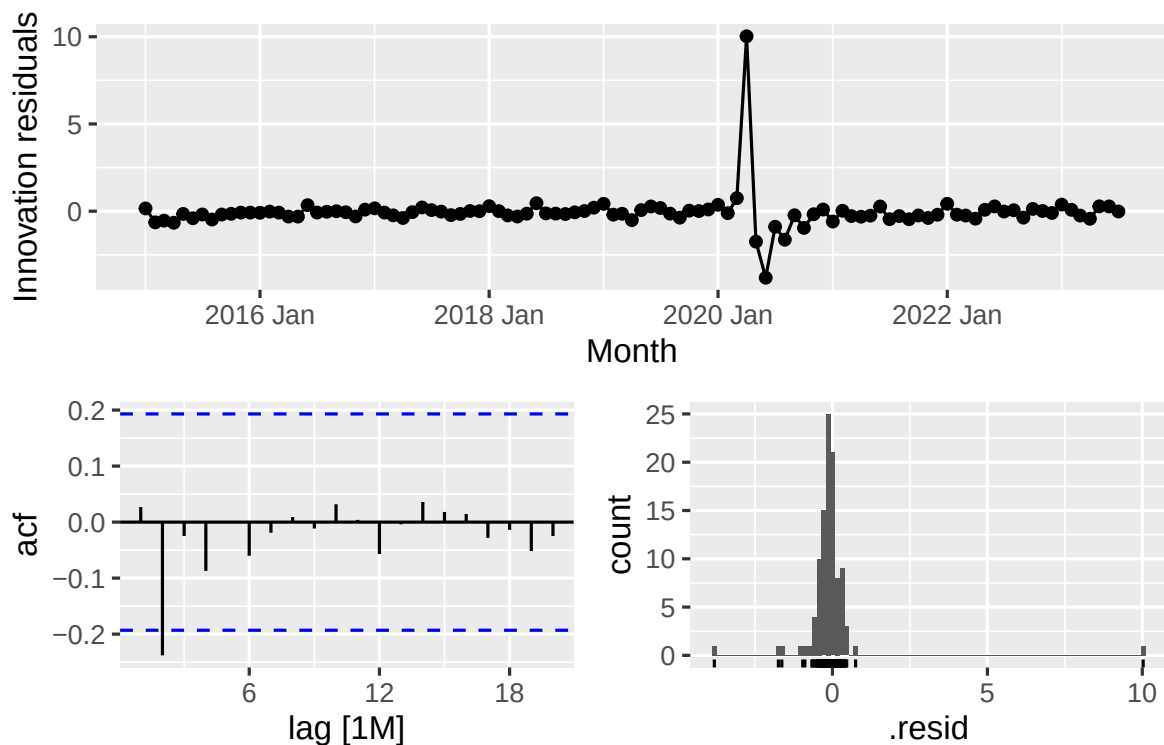
accuracy(my_fc, mydata) |>
  arrange(RMSE) |>
  select(1, 2, 4, 8:9)
#> # A tibble: 16 x 5
#>   .model          .type RMSE   MASE  RMSSE
#>   <chr>          <chr> <dbl>  <dbl>  <dbl>
#> 1 comb4          Test  0.155  0.0779 0.0569
#> 2 comb2          Test  0.180  0.0937 0.0660
#> 3 Holt_Winters_Ad_A Test  0.183  0.0949 0.0672
#> 4 Seasonal_naïve Test  0.187  0.0914 0.0686
#> 5 comb3          Test  0.190  0.103  0.0696
#> 6 Holt_Winters_Ad_M Test  0.196  0.114  0.0719
#> 7 Drift          Test  0.274  0.156  0.101
#> 8 Theta_A        Test  0.304  0.167  0.111
#> 9 Naïve          Test  0.341  0.183  0.125
#> 10 ETS_auto       Test  0.344  0.185  0.126
#> 11 comb1          Test  0.360  0.146  0.132
#> 12 SARIMA         Test  0.444  0.240  0.163
#> 13 Mean           Test  0.730  0.410  0.268
#> 14 Prophet_A      Test  1.23   0.722  0.451
#> 15 Prophet_M      Test  1.24   0.727  0.456
#> 16 N_Net          Test  2.98   1.77   1.10
```

comb4 (Holt_Winters_Ad_A + Holt_Winters_Ad_M + Theta_A) achieves the lowest RMSE, followed closely by comb2, Holt_Winters_Ad_A, and Seasonal_naïve. Among individual advanced models, Holt_Winters_Ad_A (RMSE = 0.183) and Seasonal_naïve (RMSE = 0.187) perform best. While some of the benchmark methods outperform the advanced models, the residuals in the Seasonal Naïve and Drift models did not indicate white noise. While comb4 produces the lowest raw RMSE, it is outperformed by comb2 in the residual diagnostics and Ljung-Box test. A model that passes white noise tests is preferable to one that minimizes RMSE alone.

7.1.4 Test the residuals of your preferred method

Residual diagnostics are examined for `comb2`, the ensemble of `Holt_Winters_Ad_A` and `Theta_A`. Well-behaved residuals should resemble white noise. There should be no systematic pattern over time, no significant autocorrelation in the ACF, and an approximately normal distribution. `comb2` is selected as the preferred model because it passes the Ljung-Box test ($p\text{-value} = 1.000$), indicating its residuals are fully consistent with white noise, while still achieving competitive out-of-sample accuracy.

```
my_fit |>
  select(comb2) |>
  gg_tsresiduals()
```



```
my_fit |>
  select(comb2) |>
  augment() |>
  features(.innov, ljung_box, lag = 60)
#> # A tibble: 1 x 3
```

```
#>   .model lb_stat lb_pvalue
#>   <chr>   <dbl>   <dbl>
#> 1 comb2    12.7    1.000
```

The innovation residuals plot shows values centered near zero throughout the training period, with a large spike in April–May 2020 corresponding to the COVID-19 shock. The ACF of the residuals shows only one significant spikes outside the significance bounds beyond the COVID period, indicating that the model has effectively captured the autocorrelation structure of the series. The residual histogram is approximately normal, though with a heavy right tail driven by the COVID outlier. Some heteroscedasticity is visible in the residual time plot, with variance elevated around the 2020 period.

The Ljung-Box test returns a p-value of 1.000, comfortably above the 0.05 threshold — we fail to reject the null hypothesis of no residual autocorrelation at lags 1–60. This result confirms that the comb2 residuals are consistent with white noise, and the model has extracted all available information from the training data. The COVID spike remains as a clear outlier in the residual time plot, but this reflects an unprecedented external shock rather than a modeling failure.

7.2 Forecasting

The comb2 model is now fit using the complete 129-observation dataset to produce the final forecast for the next 26 months.

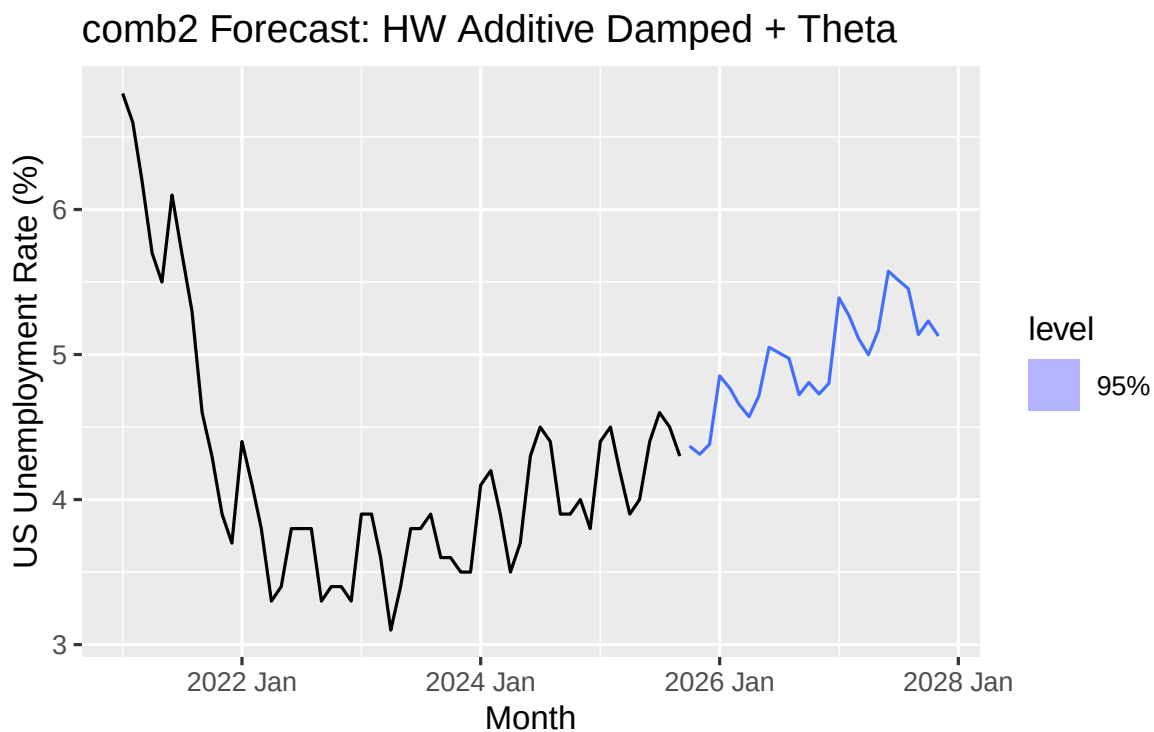
```
my_fit <- mydata |>
  model(Holt_Winters_Ad_A = ETS(box_cox(UNRATENSA, lambda) ~ error("A") + trend("Ad") +
    season("A")), Theta_A = THETA(box_cox(UNRATENSA, lambda) ~ season(period = 12))) |>
  mutate(comb2 = (Holt_Winters_Ad_A + Theta_A)/2)
```

The forecast for the next 26 months is generated and displayed below.

```
my_fit |>
  select(comb2) |>
  forecast(h = h)
#> # A fable: 26 x 4 [1M]
#> # Key:   .model [1]
#>   .model   Month UNRATENSA .mean
#>   <chr>    <mth>   <dist> <dbl>
#> 1 comb2   2025 Oct  4.368669 4.37
#> 2 comb2   2025 Nov  4.312977 4.31
#> 3 comb2   2025 Dec  4.380514 4.38
```

```
#> 4 comb2 2026 Jan 4.852854 4.85
#> 5 comb2 2026 Feb 4.767829 4.77
#> 6 comb2 2026 Mar 4.656796 4.66
#> 7 comb2 2026 Apr 4.571525 4.57
#> 8 comb2 2026 May 4.715198 4.72
#> 9 comb2 2026 Jun 5.050081 5.05
#> 10 comb2 2026 Jul 5.012032 5.01
#> # i 16 more rows
```

```
mydata |>
  filter_index("2021 Jan" ~ .) |>
  autoplot(UNRATENSA) + autolayer(mydata |>
    model(Holt_Winters_Ad_A = ETS(box_cox(UNRATENSA, lambda) ~ error("A") + trend("Ad") +
      season("A")), Theta_A = THETA(box_cox(UNRATENSA, lambda) ~ season(period = 12))) |>
    mutate(comb2 = (Holt_Winters_Ad_A + Theta_A)/2) |>
    select(comb2) |>
    forecast(h = h), level = 95) + xlab("Month") + ylab("US Unemployment Rate (%)") +
  ggtitle("comb2 Forecast: HW Additive Damped + Theta")
```



The comb2 forecast projects the U.S. unemployment rate to have a modest upward trend from approximately 4.3% in late 2025 just beyond 5.0% by late 2027, with the characteristic seasonal pattern of winter peaks and summer troughs. The stable near-term forecast with gradual upward drift is consistent with the modest upward trend observed in the unemployment rate since late 2022, and provides a meaningful and interpretable signal for stakeholders planning under uncertainty.

8 Results

The preferred model is comb2. This is the equal-weight ensemble of Holt_Winters_Ad_A and Theta_A. It is selected on the basis of both accuracy and residual diagnostics. While comb4 achieves a marginally lower RMSE, the Ljung-Box test on comb2 residuals returns a p-value of 1.000, confirming the residuals are fully consistent with white noise. A model that passes residual diagnostics is preferable for inference and interval construction, as it confirms that no information has been left out by the model.

Practically, the comb2 forecast projects a gradual upward trend in the unemployment rate from 4.3% in September 2025 toward approximately 5.0% by late 2027, consistent with the modest upward trend observed since late 2022. The 95% prediction interval widens appropriately at longer horizons, reflecting genuine uncertainty about medium-term labor market conditions.

For the Federal Reserve, a gradually rising unemployment forecast supports continued monitoring of labor market conditions, while not signaling the kind of sharp deterioration that would require emergency monetary accommodation. For businesses, the forecast implies a labor market that may become incrementally easier for hiring over the horizon, though competition for workers remains strong in the near term. For students and recent graduates, the forecast implies accessible market conditions throughout the 26-month horizon.

The distinction between statistical and practical significance is important here. The Ljung-Box result confirming white noise residuals for comb2 means the model's 95% prediction intervals are well-calibrated and can be trusted for stakeholder planning purposes, unlike models with residual autocorrelation whose intervals may be misleadingly narrow.

9 Discussion and Conclusion

This report applied 12 individual forecasting models and 4 ensemble combinations to the U.S. Monthly Unemployment Rate (UNRATENSA) for the period January 2015 through September 2025. Following the five-step forecasting process outlined in the course textbook, problem definition, data collection and preparation, model selection and evaluation, residual diagnostics, and final forecast production, the ensemble model comb2 (Holt-Winters' Additive Damped +

Theta) was identified as the preferred model based on a combination of competitive out-of-sample accuracy and passing residual diagnostics (Ljung-Box $p = 1.000$).

The comb2 ensemble succeeds by combining two complementary modeling philosophies. The Holt-Winters' Additive Damped component captures the trend and seasonal structure of the series using an exponential smoothing framework, with the damped trend preventing unrealistic long-run variation. The Theta component, anchors the forecast to recent levels. Together, they produce a forecast that is both structurally informed and empirically grounded. The final forecast projects a gradual upward drift in the unemployment rate from 4.3% toward approximately 5.0% over 26 months, with predictable seasonal variation and widening prediction intervals farther into the future.

The U.S. labor market is expected to remain broadly stable in the near term, with conditions potentially softening modestly over the two-year horizon. This signal is not a recession warning, but a reminder that the historically low unemployment rates of 2023 may not persist indefinitely.

This study is subject to several limitations worth acknowledging. First, the COVID-19 pandemic constitutes a structural break of unusual severity in the training data, producing a large residual outlier in April 2020 that does contribute to wider prediction intervals and elevated heteroscedasticity. Second, the comb2 model provides no mechanism for incorporating forward-looking information such as Federal Reserve signals, fiscal policy changes, or external economic shocks; an unexpected recession or geopolitical disruption could cause the unemployment rate to change sharply from the central forecast. Third, because the series is not seasonally adjusted, the unemployment rate is expected to rise in January and fall in spring months as a predictable seasonal pattern, independent of broader economic conditions. Stakeholders should account for this when reading month-to-month changes in the forecast. Finally, forecast accuracy at horizons beyond 12 months should be interpreted with caution, as prediction intervals grow wide and the ability of the point forecast diminishes.

Despite these limitations, this analysis provides a statistically rigorous and practically interpretable forecast of a critical labor market indicator, directly relevant to policymakers, businesses, and the broader public seeking to understand future labor market conditions.